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Fall 2026

Financial Markets

L3 – Interest Rate Risk
September 28th, 2026

comillas.edu

Summary of previous classes

Lesson 1. Financial system

Efficient-market hypothesis (Fama, 1965).

- Weak form
- Semi-strong form
- Strong form

Arbitrage concept



Lesson 2. Monetary policy

Currency

Central banks reserves

Demands deposits

Saving deposits

Large time deposits and equivalent

M0

M2

$$\frac{M2}{M0} = \text{money multiplier}$$

1928-1992

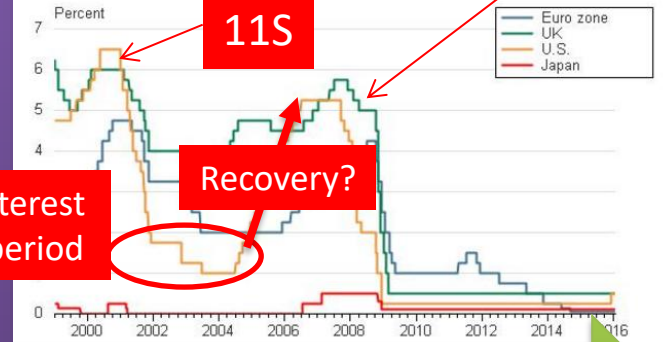
- Hyperinflation
- Bretton Woods
- Soros

16S-1992

Bank of England: 10%-15% in 1h.

2001-2008

Central bank interest rates



Low interest rates period

Recovery?

Lehman

CENTRAL BANKS: expansionary monetary policy

- Open market operations
- Standing facilities (interest rate)
- Maintenance of minimum reserves.

IS-LM Model



What are we going to do?

- We will continue working with maths...
- **Interest rate risk**
 - Bond concept: maturity, price and duration
 - Interest rate risk
 - Temporal structure of interest rates
- **Review the tools of monetary policy**
- **What happened after Lehman?**
 - Systemic risk: Iceland and Greek crisis
 - Liquidity trap
 - Quantitative easing

L3. Interest rate risk



Bond concept

What is a Bond?

IOU note (I owe you)

- The issuer: borrower
- Investor: lender
- Face value: amount borrowed
- Maturity date: term
- Coupons: fixed at issuance



I promise to pay to Capt Gilbert C. Russell or order on or before the 1st day of January next, the sum of three hundred and twenty nine dollars and fifty eight cents for value received.

\$379.58

September 27th 1860. } Merimethu Harris

Attest the Govt a seal which he is to leave at Talbot in his own hand.

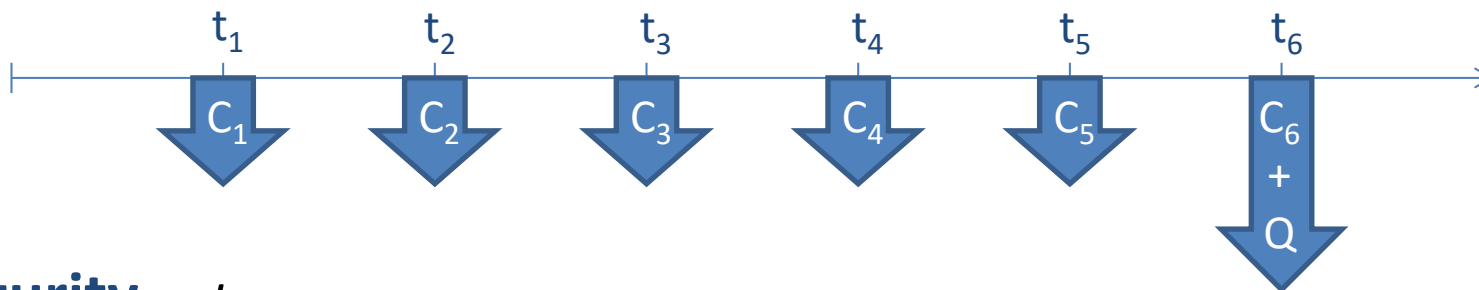
The Malden & Melrose Railroad Co. .
[\$15] Will pay the Bearer, at the Office of their Treasurer, in Boston, FIFTEEN DOLLARS, on the 1st day of April, 1863, for six months' interest on Bond No. 2, dated C. 1st, 1860.
C. H. Powers Treasurer.



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Maturity, price and duration



Maturity = t_6

Price = $VA(C_1) + VA(C_2) + \dots + VA(C_6 + Q) = \frac{C_1}{(1+r)^{t_1}} + \frac{C_2}{(1+r)^{t_2}} + \dots + \frac{C_6 + Q}{(1+r)^{t_6}}$

Duration = $\frac{t_1 \cdot VA(C_1) + t_2 \cdot VA(C_2) + \dots + t_6 \cdot VA(C_6 + Q)}{VA(C_1) + VA(C_2) + \dots + VA(C_6 + Q)} = \frac{t_1 \cdot \frac{C_1}{(1+r)^{t_1}} + t_2 \cdot \frac{C_2}{(1+r)^{t_2}} + \dots + t_6 \cdot \frac{C_6 + Q}{(1+r)^{t_6}}}{\frac{C_1}{(1+r)^{t_1}} + \frac{C_2}{(1+r)^{t_2}} + \dots + \frac{C_6 + Q}{(1+r)^{t_6}}}$



Maturity, price and duration

Calculate duration of following cash flow using a discount of 15%:

Tiempo	1	2	3
Flujo de caja	3.000,00	2.000,00	1.000,00

$$duration = \frac{1 \cdot \frac{3.000}{1,15} + 2 \cdot \frac{2.000}{1,15^2} + 3 \cdot \frac{1.000}{1,15^3}}{\frac{3.000}{1,15} + \frac{2.000}{1,15^2} + \frac{1.000}{1,15^3}} = 1,59 \text{ years}$$

Calculate duration of a **zero coupon bond** with maturity of 10 years and a **yield to maturity** of 5%. Also calculate the issue price and the **discount**. (Nominal=1.000€)

$$duration = maturity = 10 \text{ years}$$

$$Price = \frac{1.000}{1,05^{10}} = 613,91$$

$$Discount = \frac{1.000 - 613,91}{1.000} = 38,61\%$$



Maturity, price and duration

A Bond is issue without discount, with maturity of 3 years, facial of 1.000€ and three coupons of 100€ to be paid yearly. Calculate yield to maturity and duration (with IRR)

Time		1	2	3
Debt	- 1.000,00			1.000,00
Coupon		100,00	100,00	100,00
Cash Flow	- 1.000,00	100,00	100,00	1.100,00

$$duration = \frac{1 \cdot \frac{100}{1,1} + 2 \cdot \frac{100}{1,1^2} + 3 \cdot \frac{1.100}{1,1^3}}{\frac{100}{1,1} + \frac{100}{1,1^2} + \frac{1.100}{1,1^3}} = 2,74 \text{ years}$$

$$IRR \Rightarrow -1.000 + \frac{100}{1+r} + \frac{100}{(1+r)^2} + \frac{1.100}{(1+r)^3} = 0 \Rightarrow IRR = 10\% \Rightarrow \frac{100}{1,1} + \frac{100}{1,1^2} + \frac{1.100}{1,1^3} = 1.000$$

A Bond is issue without discount, with maturity of 10 years, facial of 1.000€ and ten coupons of 100€ to be paid yearly. Calculate yield to maturity and duration (with IRR)

Time		1	2	3	4	5	6	7	8	9	10
Debt	- 1.000,00										1.000,00
Coupon		100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00
Cash flow	- 1.000,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	100,00	1.100,00

	Total										
NPV	1.000,00	90,91	82,64	75,13	68,30	62,09	56,45	51,32	46,65	42,41	424,10
time x NPV	6.759,02	90,91	165,29	225,39	273,21	310,46	338,68	359,21	373,21	381,69	4.240,98
Duration	6,76										



Interest rate risk

A government Bond is issued without discount, with maturity of 10 years, face of 1.000€ and ten coupons of 20€ to be paid yearly. The interest rate is 2%

$$IRR \Rightarrow -1.000 + \frac{20}{1+r} + \frac{20}{(1+r)^2} + \dots + \frac{1.020}{(1+r)^{10}} = 0 \Rightarrow IRR = 2\%$$

If the bond is sold after one year time and the interest rate remains unchanged the return is:

$$Value = \frac{20}{1,02} + \frac{20}{1,02^2} + \dots + \frac{1.020}{1,02^9} = 1.000$$

$$\text{return} = \frac{1.000 + 20 - 1.000}{1.000} = 2\%$$

i=2%

...but... if interest rate rises to 3% or falls to 1%

$$Value = \frac{20}{1,03} + \frac{20}{1,03^2} + \dots + \frac{1.020}{1,03^9} = 922,14$$

$$\text{return} = \frac{922,14 + 20 - 1.000}{1.000} = -5,79\%$$

i=3%

$$Value = \frac{20}{1,01} + \frac{20}{1,01^2} + \dots + \frac{1.020}{1,01^9} = 1.085,66$$

$$\text{return} = \frac{1.085,66 + 20 - 1.000}{1.000} = 10,57\%$$

i=1%

Interest rate risk

The interest rate risk can affect investment in two ways:

- **Changes the market value** of assets whose cash flows are independent of the interest rate .
- Changes the **cash flow's values** of those assets whose cash flows depend on the interest rate.

Liabilities with fixed rate vs.

(Market value changes and not cash flows)

If a firm's ability to sell its product is affected by the level of real interest rates, then the firm will want to minimize the exposure of its liabilities to interest rate changes – e.g. fixed rate liabilities

Higher rates: higher financial expenses and lower revenues
Lower rates: lower financial expenses and higher revenues

Liabilities with floating rates

(Cash flows change and not market value)

If changes in interest rates mainly reflect changes in inflation and if a firm's operating profits increase with inflation, then the firm will want its liabilities to be exposed to interest rate risk – e.g. floating rate liabilities

Higher rates: higher financial expenses and higher revenues
Lower rates: lower financial expenses and lower revenues

Modified duration. Quantifies the measurable change in the value of a security in response to a change in interest rates.

$$\text{Modified duration} = \frac{\text{Duration}}{1 + r} \%$$

- Rate risk exposure measure
- High duration = high rate risk impact.
(Highest duration: zero coupon)
- Interest rise price falls

Term Structure of Interest Rates (TSIR)

What is the interest rate?

- EURIBOR, Central Bank Interest rate, the Yield curve, Kd?
- Who controls interests rates
- What about monetary policy?

Public finance

Government budget - (presupuestos generales del estado)

Primary market

El Tesoro anuncia los bonos y obligaciones que se subastarán el día 18 de diciembre

Detalle:
El Tesoro anuncia los Bonos y Obligaciones que se subastarán el día 18 de diciembre.
Fecha de subasta: 18 de diciembre.

Obligaciones del Estado cupón 4,60%, Vto.: 30 julio 2019
Intereses corrientes: 1,84 %
Obligaciones del Estado cupón 4,40%, Vto.: 31 octubre 2023
Intereses corrientes: 0,64 %
Obligaciones del Estado cupón 2,75%, Vto.: 31 octubre 2024
Intereses corrientes: 0,40 %

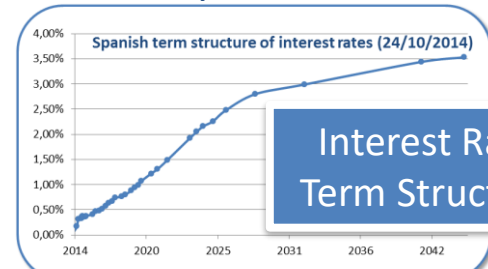
El precio que se solicite en las ofertas competitivas se formulará ex cupón.



Secondary market

EMISIÓN	Tipo	Cupón	Vencimiento	Rendimiento anual medio	Número de operaciones	Medio	Máximo	Mínimo
ES001411211	Letra del Estado	CUP-0	21/11/2014	0,06%	1	99,996	99,996	99,996
ES0000012371	Bono del Estado	2,75%	31/03/2015	0,32%	1	101,030	101,030	101,030
ES001510160	Letra del Estado	CUP-0	16/10/2015	0,36%	4	99,653	99,653	99,651
ES0000012399	Bono del Estado	3,75%	31/10/2015	0,38%	2	103,380	103,380	103,375
ES0000012205	Bono del Estado	3,25%	30/04/2016	0,41%	7	104,241	104,245	104,210
ES0000012018	Obligación del Estado	3,80%	31/01/2017	0,52%	1	107,339	107,339	107,339
ES0000012389	Obligación del Estado	5,50%	30/04/2021	1,31%	19	125,961	126,000	125,900
ES0000012409	Obligación del Estado	3,80%	30/04/2024	2,05%	10	115,001	115,080	114,940
ES0000012682	Obligación del Estado	2,75%	31/10/2024	2,16%	19	105,305	105,380	105,100
ES0000012425	Obligación del Estado	5,15%	31/10/2028	2,80%	5	126,867	127,000	126,620
ES0000012411	Obligación del Estado	5,75%	30/07/2032	2,99%	3	137,572	137,901	137,570
ES0000012157	Obligación del Estado	4,70%	30/07/2041	3,44%	1	121,880	121,880	121,880
ES0000012444	Obligación del Estado	5,15%	31/10/2044	3,53%	4	129,738	129,770	129,700

...each day...



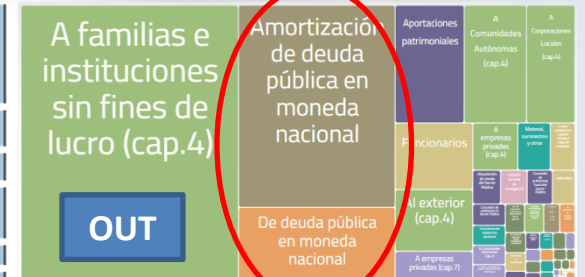
Interest Rate
Term Structure

<http://www.civio.es/>

<http://dondevanmisimpuestos.es/resumen>

At least, with one year time:

Governments need to know, exactly what to pay, what to get and when.



<http://dondevanmisimpuestos.es/politicas>



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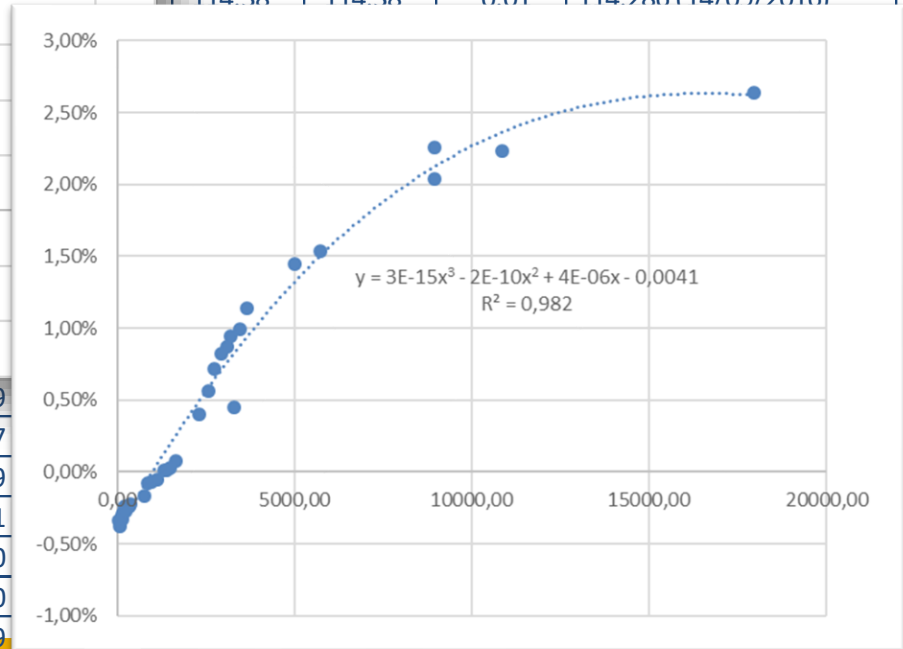
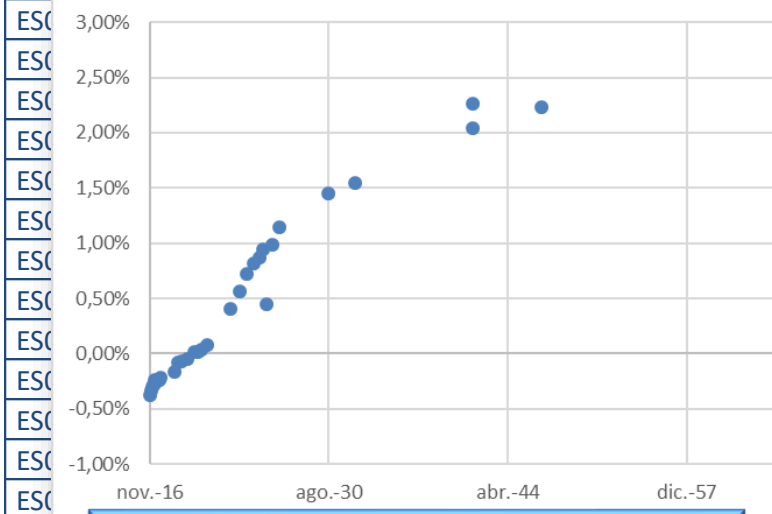
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Term Structure of Interest Rates (TSIR).

<http://www.bde.es/webbde/es/secciones/informes/banota/boletin.html>

EMISION	NÚMERO OPERACS	IMPORTE CONTRAT	PRECIO (EX-CUPON)			RENTDO. INTERNO	ANTERIOR PRECIO MEDIO
			MEDIO	MAXIMO	MINIMO		
ES00000128E2 O EST 3.45 30.07.66	2	142,02	122,28	125,88	118,69	2,64	119,059 (15/09/2016)
ES00000124I2 B EST 2.10 30.04.17	1	11,00	101,45	101,4			
ES00000124B7 B EST 3.75 31.10.18	1	0,07	108,33	108,3			
ES00000124B7 B EST 3.75 31.10.18			100,7	107,3			
ES00000124B7 B EST 3.75 31.10.18			113,60	113,60		0,05	113,362 (15/09/2016)
ES00000124B7 B EST 3.75 31.10.18			114,38	114,38		0,01	114,286 (14/09/2016)

Features of government treasure TSIR:
 - no credit risk (in theory)
 - no liquidity problems



Letras del Estado (Zero coupon)
 Maturity: 3, 6, 12 and 18 months

Bonos del Estado (Yearly coupon)
 Maturity: 3 and 5 years

Obligaciones del Estado (yearly coupon)
 Maturity: 10, 15 and 30 years.

Term Structure of Interest Rates (TSIR).



United Kingdom
Debt Management
Office

<http://www.dmo.gov.uk/>

<http://www.bankofengland.co.uk/statistics/pages/yieldcurve/default.aspx>

GILT MARKET GILT REFERENCE PRICES LATEST AVAILABLE PRICES									
Publication of these prices does not constitute an offer to buy or sell securities. To export the information from this report click on one of the buttons at the end of this page.									
Gilt Name	ISIN Code	Redemption Date	Close of Business Date	Clean Price (£)	Dirty Price (£)	Accrued Interest (£)	Yield (%)	Modified Duration	
Conventional									
9 1/2% Treasury Stock 2017	GB0008601148	25-Aug-2017	03-Feb-2017	104.860000	108.783234	3.923234	-0.057478	0.53	
1% Treasury Gilt 2017	GB00B7F95958	07-Sep-2017	03-Feb-2017	100.550000	100.969890	0.419890	0.051825	0.58	
3% Treasury Gilt 2018	GB00B7VWPC84	07-Mar-2018	03-Feb-2017	105.300000	107.390448	2.090448	0.088553	1.04	
1 1/2% Treasury Gilt 2018	GB00B8P8H444	22-Jun-2018	03-Feb-2017	101.080000	101.731790	0.651790	0.067079	1.45	
4 1/2% Treasury Gilt 2019	GB00B39R3F84	07-Mar-2019	03-Feb-2017	109.100000	110.989503	1.889503	0.118460	1.98	
1 1/2% Treasury Gilt 2019	GB00B0V0F150	22-Jun-2019	03-Feb-2017	103.760000	103.852514	0.072514	0.203785	2.41	
9 1/2% Treasury Gilt 2020	GB00B4H8F441	07-Sep-2019	03-Feb-2017	109.200000	110.774585	1.574585	0.174623	2.45	
4 1/2% Treasury Stock 2020	GB00B0960C55	07-Mar-2020	03-Feb-2017	113.730000	115.724475	1.994475	0.270574	2.85	
2% Treasury Gilt 2020	GB00BN55R189	22-Jun-2020	03-Feb-2017	105.470000	105.552873	0.082873	0.405872	3.35	



BANK OF CANADA

<https://www.bankofcanada.ca/>

<https://www.bankofcanada.ca/rates/interest-rates/bond-yield-curves/>



U.S. DEPARTMENT OF THE TREASURY

<https://www.treasury.gov/resource-center/data-chart-center/interest-rates/>



中國銀行
BANK OF CHINA

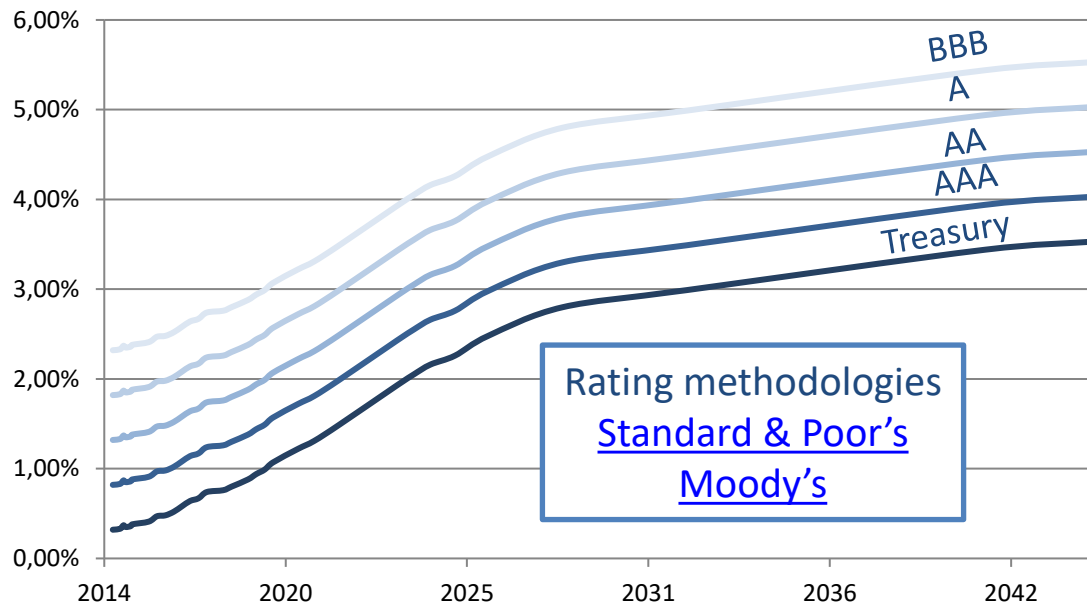
<https://www.boc.cn/en/index.Html>



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Term Structure of Interest Rates (TSIR).

There is a different TSIR each day, and not all kind of assets fit in the same TSIR.

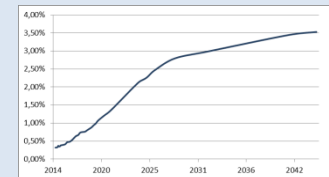


Dynamic yield – curve vs SP500

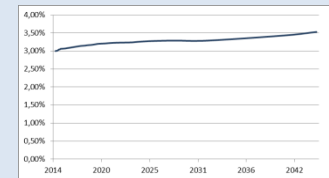
<http://stockcharts.com/freecharts/yieldcurve.php>

Types of curves

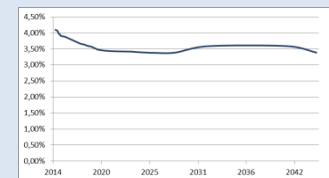
Normal curve (increasing)



Flat curve



Decreasing



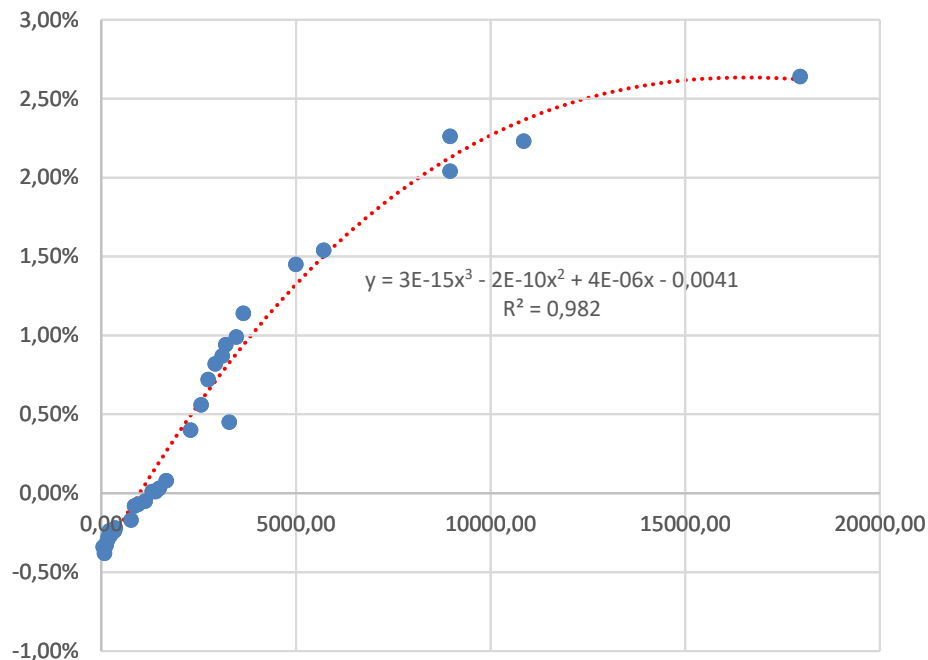
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Term Structure of Interest Rates (TSIR).

Zero Coupon curve (I)

Using excel, first “scatter plot” and then “polynomic trend line”, we get the polynomic approximation to the TSIR.



With the formula we can get following table:

Years	IRR
1	-0,27%
2	-0,13%
3	0,00%
4	0,13%
5	0,25%
6	0,36%
7	0,48%
8	0,58%
9	0,69%
10	0,78%



Term Structure of Interest Rates (TSIR).

Zero Coupon curve (II)

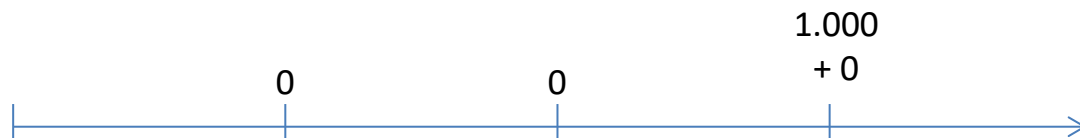
A bond is equivalent to a surface of zero coupon bonds with same payments.

$$IRR_{1\text{ year}} = -0,27\% \rightarrow ZCC_{1\text{ year}} = -0,27\%$$

$$IRR_{2\text{ year}} = -0,13\%$$

$$IRR_{3\text{ year}} = 0,00\%$$

$$1.000 = \frac{-1,3}{0,9973} + \frac{998,68}{(1+ZCC_2)^2} \rightarrow ZCC_2 = -0,13\%$$



$$1.000 = \frac{0}{0,9973} + \frac{0}{(1+ZCC_2)^2} + \frac{1.000}{(1+ZCC_3)^3} \rightarrow ZCC_3 = 0,00\%$$

$$IRR_3 = 5\%$$



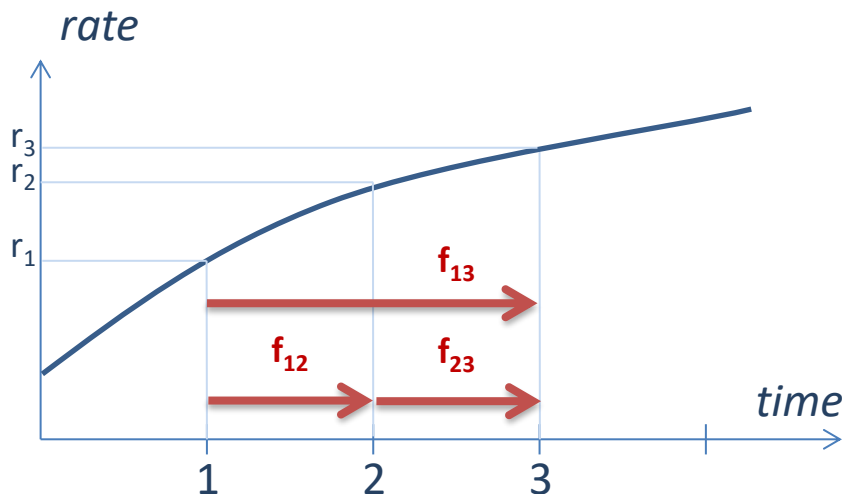
$$1.000 = \frac{50}{0,9973} + \frac{50}{(1+ZCC_2)^2} + \frac{1.050}{(1+ZCC_3)^3} \rightarrow ZCC_3$$

Year	Zero CC	IRR
1	-0,27%	-0,27%
2	-0,13%	-0,13%
3	0,00%	0,00%
4	0,13%	0,13%
5	0,25%	0,25%
6	0,37%	0,36%
7	0,48%	0,48%
8	0,59%	0,58%
9	0,70%	0,69%
10	0,80%	0,78%



Term Structure of Interest Rates (TSIR).

Spot rate and forward rate



Spot rate: r_1 , r_2 and r_3

Forward rate: f_{12} , f_{23} and f_{13}

$$(1 + r_2)^2 = (1 + r_1)(1 + f_{12})$$

$$(1 + r_3)^3 = (1 + r_2)^2(1 + f_{23}) = (1 + r_1)(1 + f_{12})(1 + f_{23})$$

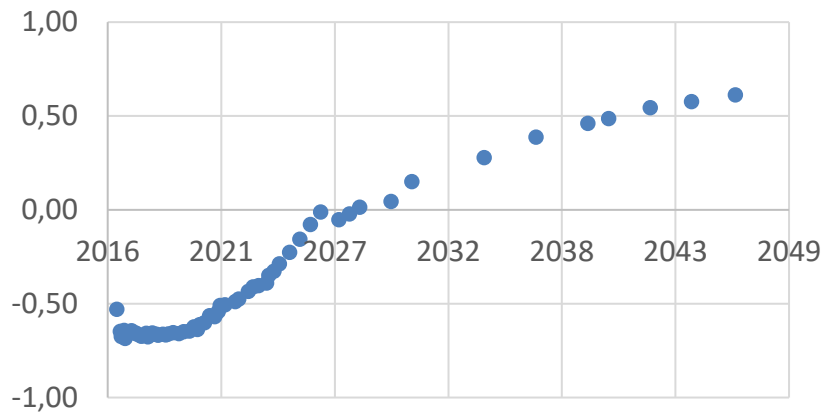
$$(1 + r_3)^3 = (1 + r_1)(1 + f_{13})^2$$



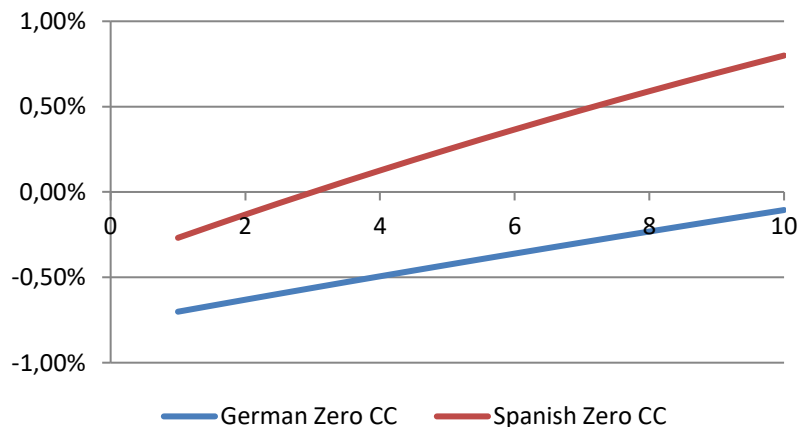
Term Structure of Interest Rates (TSIR).

German federal securities

http://www.bundesbank.de/Navigation/EN/Service/Federal_securities/Prices_and_yields/prices_and_yields.html



Year	German Zero CC	Spanish Zero CC	Difference
1	-0,70%	-0,27%	0,43%
2	-0,63%	-0,13%	0,50%
3	-0,56%	0,00%	0,56%
4	-0,49%	0,13%	0,62%
5	-0,43%	0,25%	0,68%
6	-0,36%	0,37%	0,73%
7	-0,30%	0,48%	0,78%
8	-0,23%	0,59%	0,82%
9	-0,17%	0,70%	0,86%
10	-0,11%	0,80%	0,90%



Tools of monetary policy

- **Remembering lesson 2: What are the three tools ECB can use?**
 - standing facilities,
 - minimum reserve requirements for credit institutions,
 - open market operations.

Is it really useful now?
What is the aim of the ECB?
Let see the evolution of the financial system...

Open market operations.	Transaction type		Maturity	Frequency	Procedure
	Liquidity injection	liquidity absorption			
Main refinancing operations	Temporary operations		2 weeks	Weekly	Standard tenders
Longer-term refinancing operations	Temporary operations		3 months	Monthly	Standard tenders
Fine-tuning operations	Temporary operations Currency Swaps	Temporary operations Currency Swaps Collection of deposits	No standardized	No regular	Fast tenders Bilateral procedures
	Securities purchase	Sale of securities		No regular	Bilateral procedures
Structural operations	Temporary operations	Issuance of debt certificates	Standardized / no st	Regular and n	Standard tenders
	Securities purchase	Sale of securities		No regular	Bilateral procedures



Evolution of the financial system

We talked in previous classes about:

- 1921-1924 German hyperinflation
- 1944-1971 Bretton Woods. Gold standard (fixed exchange rates). Stability
- 1979-1998 European Monetary System. 1992 Soros and the pound.
- 2001 The Federal Reserve's Response to the Sept. 11 Attacks
- 2001-2008 Low interest rate period with the “recovery” from 2006.
- **Sept 2008 – Lehman Brothers Bankruptcy**

...and then?

Systemic risk
Zero interest rate policy

Iceland crisis
Greek crisis

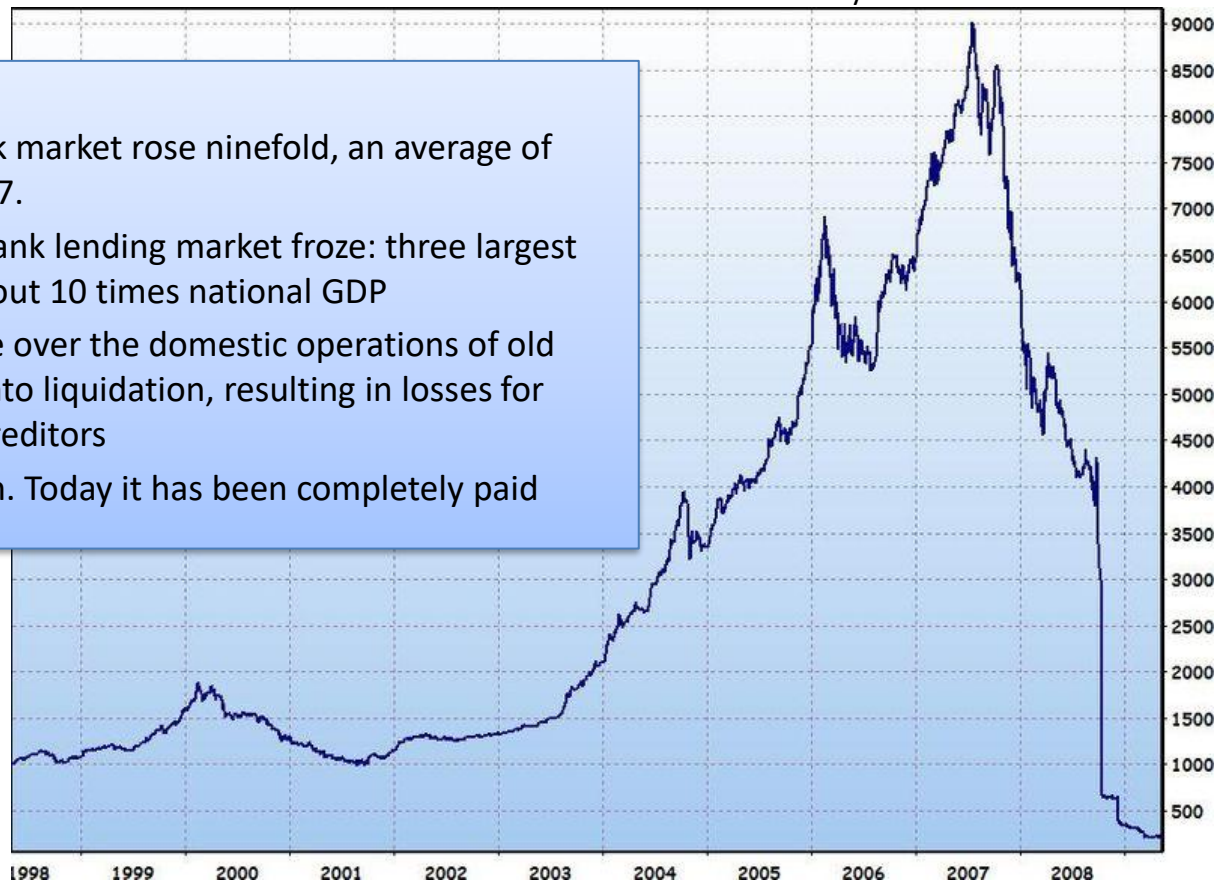
Liquidity trap
Quantitative easing



Systemic risk: Iceland crisis

OMX Iceland 15 from January 1998 to October 2008

- 2000 Banks were privatized.
- Banks borrowed heavily. Ej: stock market rose ninefold, an average of 44% each year from 2001 to 2007.
- After Lehman bankruptcy interbank lending market froze: three largest Banks had liabilities equaling about 10 times national GDP
- New banks were founded to take over the domestic operations of old banks. The old banks were put into liquidation, resulting in losses for their shareholders and foreign creditors
- IMF agreed to lend €1.875billion. Today it has been completely paid



Systemic risk: Greek crisis

- Officially started in 2009
- Causes:
 - Global recession
 - Structural weaknesses of the Greek economy
 - Government debt levels and deficits were undercounted by the Greek government.
- Crisis of confidence & CDS

As previously happened with mortgages, different kind of debts were packaged and sold in **CDOs (Collateralized Debt Obligation)** and the credit risk was assured buying a **CDS (Credit Default Swaps)**

Greece Paid Goldman \$300 Million To Help It Hide Its Ballooning Debts

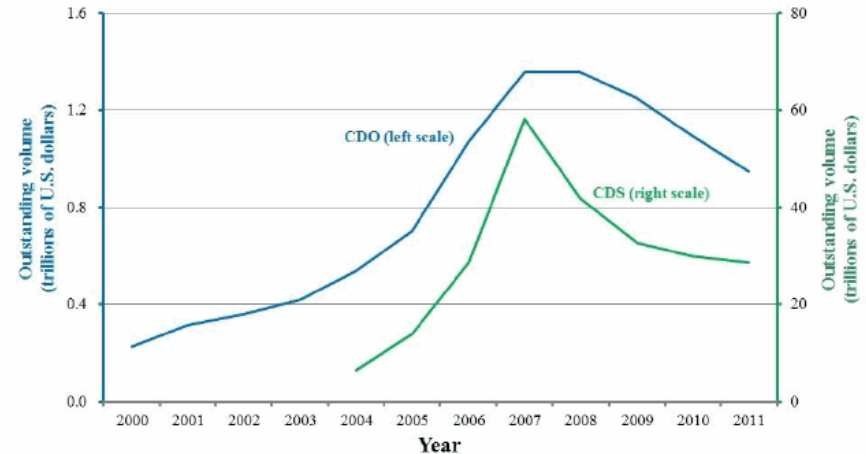
[Business insider Feb 2010](#)



Henry Blodget

Feb. 14, 2010, 9:27 AM ▲ 21,444 ○ 59

Growth in CDO and CDS Markets, 2000-11

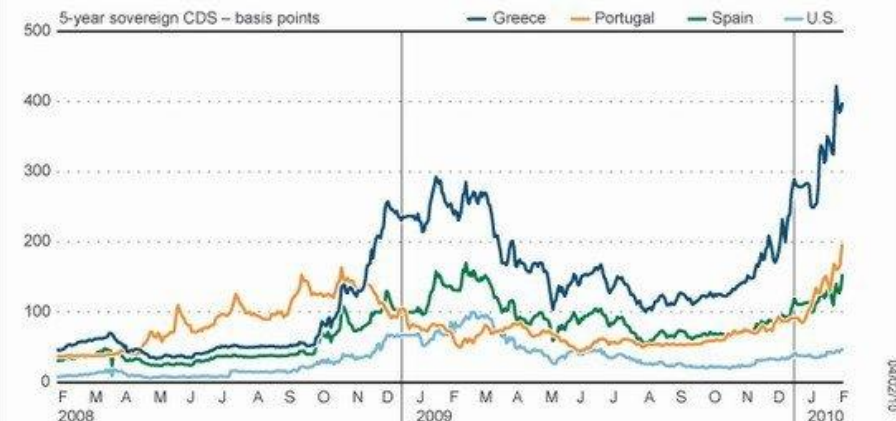


Note: CDO is collateralized debt obligation; CDS is credit default swap. CDS data start in 2004.

Source: For CDO, Securities Industry and Financial Markets Association; for CDS, Bank for International Settlements.

Global sovereign credit default swaps

The cost of insuring sovereign debt against default is rising, particularly in Europe



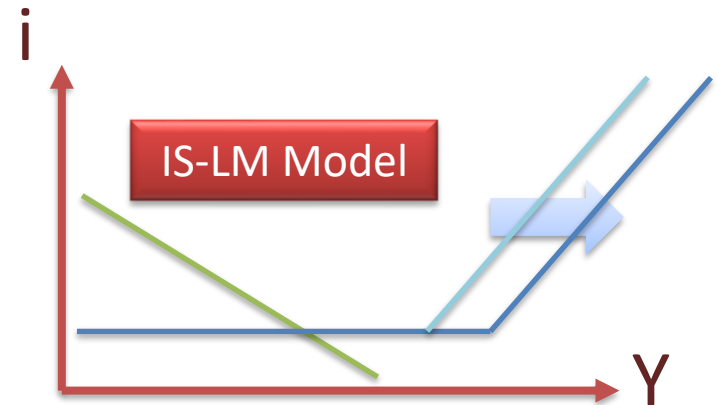
Source: Thomson Reuters Datastream

Reuters graphic/Stephen Culp

Zero interest rates policy: Liquidity trap

- Low interest rates and high savings rates
- Investors choose to avoid bonds and keep their funds in savings, because of the prevailing belief that interest rates will soon rise.
- **Traditional monetary policy doesn't work**

If we think that interest rates are going to rise
¿what would happen if we buy public debt?



We have two problems:

- Traditional monetary policy doesn't work
- People don't want to buy public debt

¿Solution? **quantitative easing**



Quantitative easing

Expanded Asset Purchase Programme (APP) [link](#)

- third covered bond purchase programme (CBPP3)
- asset-backed securities purchase programme (ABSPP)
- public sector purchase programme (PSPP)
- corporate sector purchase programme (CSPP)

How does monetary policy work?

Monthly purchases in public and private sector securities amount to €80 billion on average

Eurosystem holdings under the expanded asset purchase programme

What are the risks?

Changes of holdings (last two months)	ABSPP	CBPP3	CSPP	PSPP	APP
Holdings* July 2016	20,368	186,634	13,214	944,852	1,165,068
Monthly net purchases	-226	3,504	6,707	50,513	60,498
Quarter-end amortisation adjustment	0	0	0	0	0
Holdings* August 2016	20,142	190,139	19,921	995,364	1,225,566

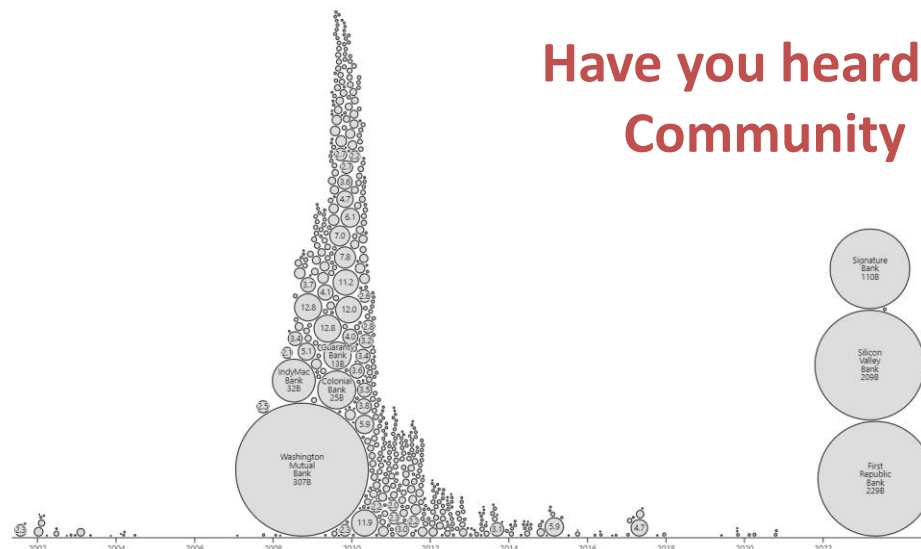
*At amortized cost, in euro million, at month end.

Figures may not add up due to rounding. Figures are preliminary and may be subject to revision.



Silicon Valley Bank Crisis

- High exposure to bonds with a maturity of over 10 years.
- The FED raises interest rates by more than 400 basis points in less than a year.
- After the pandemic, technological clients withdraw deposits to cover layoffs and operational expenses.
- Significant losses occur due to the depreciation of long-term bonds held on the balance sheet.
- The FED intervenes to prevent contagion and stabilize the financial system.



**Have you heard of New York
Community Bancorp?**

<https://observablehq.com/@mbostock/bank-failures>



Past, present and Future

1929 financial crisis

- The 1929 crisis was a financial crisis. As the U.S. government did not intervene, it turned into the Great Depression, lasting for years.

2008 financial crisis

- The 2008 crisis was also a financial crisis. However, immediate government intervention prevented it from becoming an economic depression.
- Ben Bernanke, then Chairman of the FED, had studied the 1929 crisis and implemented swift measures, which earned him the Nobel Prize in Economics.

Post-SVB crisis management

- Bank crises after SVB in March 2023 occur in real-time, and the FED manages these crises in real-time, often without public awareness.

What we have done until now?

Lesson 1.

Introduction

Spanish financial
sector

Function of Financial Markets and financial intermediaries

Financial Market instruments

Understanding financial markets through the Financial Crisis
(Subprime)

An overview of Financial Risks

Evolution of the financial system

Lesson 2.

EMU and
monetary policy

Central Banks: origins, structures and functions

The European Central Bank (ECB) and the Monetary policy

The creation of money: What is money?

Supply and demand: monetary base

Monetary policy

Lesson 3.

Interest rate risk
and money
markets

Concepts and classes

Understanding interest rates

The yield curve (risk and term structure)

Interbank market

Commercial paper and repos

Interest rates



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Thanks

